

MVLU COLLEGE

AIM: Implementation of DSR Routing Protocols

```
...
(file "dsr_sim.tcl" line 1)
mvlu@IT-09:~$ nano dsr_sim.tcl
mvlu@IT-09:~$ cat dsr_sim.tcl
set ns [new Simulator]
$ns node-config -adhocRouting DSR \
-llType LI \
-macType Mac/802.11 \
-ifqType Queue/DropTail/PriQueue \
-ifqLen 50 \
-antType Antenna/OmniAntenna \
-propType Propagation/TwoRayGround \
-phyType Phy/WirelessPhy \
-channelType Channel/WirelessChannel \
-topoInstance $topo \
-agentTrace ON \
-routerTrace ON \
-macTrace OFF
mvlu@IT-09:~$ nano dsr_sim.tcl
mvlu@IT-09:~$ ls
aodv_out.nam  aodv_sim.tcl  dsr.tr        dsr_sim.tcl      nam_1.15-6.dsc  nam_error.txt
aodv_out.tr   dsr.nam       dsr_practical  nam_1.15-6.debian.tar.xz  nam_1.15-orig.tar.gz  'nano aodv_sim.tcl'
mvlu@IT-09:~$ nano dsr_out.nam
nm:
mvlu@IT-09:~$ nano dsr.nam
nm:
mvlu@IT-09:~$ wc -l dsr.nam
0 dsr.nam
mvlu@IT-09:~$ head -20 dsr.nam
mvlu@IT-09:~$ nano dsr_sim.tcl
mvlu@IT-09:~$ rm -f dsr.nam dsr.tr
mvlu@IT-09:~$ rm dsr_sim.tcl
warning: Please use -channel as shown in tcl/ex/wireless-wmf.tcl
can't read "phyType": no such variable
while executing
"while executing
$node add-interface $chan $propInstance_ $llType_ $macType_ $ifqType_ $ifqLen_ $phyType_ $antType_ $topoInstance_ $inerrProc_ $outerrProc_ $FECProc_"
(procedure "o3" line 83)
(Simulator create-wireless-node line 83)
invoked from within
"_o3 create-wireless-node"
("eval" body line 1)
invoked from within
```

```
GNU nano 7.2 dsr_sim.tcl
set ns [new Simulator]

set tracefd [open dsr.tr w]
set namfd [open dsr.nam w]

set topo [new Topography]
$topo load_flatgrid 500 500

set god_ [create-god 1]

$ns trace-all $tracefd
$ns namtrace-all-wireless $namfd 500 500

set chan_1_ [new Channel/WirelessChannel]

$ns node-config \
-adhocRouting DSR \
-llType LI \
-macType Mac/802.11 \
-ifqType Queue/DropTail/PriQueue \
-ifqLen 50 \
-antType Antenna/OmniAntenna \
-propType Propagation/TwoRayGround \
-phyType Phy/WirelessPhy \
-channel $chan_1_ \
-topoInstance $topo \
-agentTrace ON \
-routerTrace ON \
-macTrace OFF

set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

$n0 random-motion 0
$n1 random-motion 0
$n2 random-motion 0
```

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```
GNU nano 7.2 dsr.sim.tcl
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

$n0 random-motion 0
$n1 random-motion 0
$n2 random-motion 0

$ns initial_node_pos $n0 50
$ns initial_node_pos $n1 150
$ns initial_node_pos $n2 250

set udp [new Agent/UDP]
$ns attach-agent $n0 $udp

set null [new Agent/Null]
$ns attach-agent $n2 $null

$ns connect $udp $null

set cbr [new Application/Traffic/CBR]
$cbr set packetSize_ 512
$cbr set interval_ 0.1
$cbr attach-agent $udp

$ns at 1.0 "$cbr start"
$ns at 8.0 "$cbr stop"

proc finish {} {
    global ns tracefd namfd
    $ns flush-trace
    close $tracefd
    close $namfd
    exit 0
}

$ns at 10.0 "finish"
```

```
Invoked from within
"God instance"
(procedure ".o13" line 113)
(Object next line 113)
Invoked from within
".o13 next .o11 .o12 LL Mac/802.11 Queue/DropTail/PrsQueue 50 Phy/WirelessPhy Antenna/OmiAntenna .o10 {} {}"
("eval" body line 1)
Invoked from within
"eval $self next $args"
(procedure ".o11" line 2)
(Simulator add-interface line 2)
Invoked from within
"$node add-interface $chan $propInstance_ $lltype_ $macType_ $ifqType_ $ifqlen_ $phyType_ $antType_ $topoInstance_ $inerrProc_ $outerrProc_ $FECPProc_"
(procedure ".o1" line 83)
(Simulator create-wireless-node line 83)
Invoked from within
".o3 create-wireless-node"
("eval" body line 1)
Invoked from within
"eval $self create-wireless-node $args"
(procedure ".o3" line 23)
(Simulator node line 23)
Invoked from within
"$ns node"
Invoked from within
"set n0 [$ns node]"
(file "dsr.sim.tcl" line 20)

mvlu@IT-09:~$ nano dsr.sim.tcl
mvlu@IT-09:~$ rm -f dsr.nam dsr.tr
ns dsr.sim.tcl
num_nodes is set 3
INITIALIZE THE LIST xlistHead
channel.cc:sendUp = Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
mvlu@IT-09:~$ wc -l dsr.nam
510 dsr.nam
mvlu@IT-09:~$ nam dsr.nam
nam:
mvlu@IT-09:~$ which nam
/usr/bin/nam
mvlu@IT-09:~$ nam -v
```